Who May Overrule the Agent?
Evidence-Gated Authority in LLM Review Institutions

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Abstract

Multi-agent language-model systems often add a reviewer to improve an upstream agent's decision. Yet a reviewer can both repair errors and replace decisions that were already correct. We study this as a problem of institutional jurisdiction: conditional on the same proposal, task information, model, and review output, when is the reviewer authorized to make its recommendation binding? We compare broad override authority with an evidence gate that permits replacement only after the reviewer produces a mechanically valid record of a binding-constraint violation and a compliant repair. Across two locally run open-weight models, 96 spatial-policy tasks, and a 96-task portfolio-transfer benchmark, the gate sharply reduces corruption of correct proposals while retaining most repairs of violations. The directions of this tradeoff are partly implied by the rule: the gate preserves nonviolating proposals, including both correct and suboptimal ones. The empirical quantities are its magnitude, the repairs lost to invalid certificates, and the reviewer transition rates that determine deployment outcomes. Consequently, the institution that maximizes exact optimality depends on the prevalence of upstream proposal states and on the reviewer's state-conditional competence. On naturally generated spatial proposals, the gate improves compliance for both models but its exact-optimality effect differs by model. The result is not that restricted review is universally better. It is that reviewer competence alone is insufficient for institutional design: the scope of authorized intervention determines both the correction domain and the corruption surface.

1 Introduction

A common agent architecture delegates a task to one language model and asks another model to critique, verify, or revise the result. The second call is normally treated as an additional capability: if review quality improves, system quality should improve. This view leaves an institutional choice implicit. A reviewer may be permitted to advise, to reject only on specified grounds, or to replace any decision. These rules can produce different outcomes even when the reviewer emits exactly the same recommendation.

A recent real-world incident illustrates why the surrounding institution matters. METR's independent investigation of the OpenAI/Hugging Face incident reported that agents intended to operate in isolation discovered a shared communication channel, adopted assignment, ownership, hold, veto, and stop conventions, and sometimes risked their individual task success for collective projects [21]. The incident does not identify the mechanism tested here, and our experiment does not reproduce it. It motivates treating agent coordination and authority rules as safety-relevant objects in their own right.

That distinction matters because language-model review is not monotone. Models can identify defects and improve outputs [12, 19], but unguided reconsideration can also degrade correct reasoning [8]. Debate and multi-agent voting can improve benchmark performance [5, 3, 24], while recent work finds that voting often explains more of the gain than debate itself [4]. In homogeneous teams, unguided peer exchange can destabilize initially correct reasoning, while plurality consensus can discard correct answers already present in the generation pool [2]. Correct minority information and blind conformity remain live failure modes [7, 22]. The unresolved design question is therefore not only whether reviewers are capable, but when their judgments acquire force.

We isolate that question in a two-agent decision institution. An upstream agent selects a binding action from seven alternatives under two ordered duties: first satisfy a protected constraint; then optimize aggregate welfare among eligible alternatives. A reviewer sees the same task and proposal. Under *broad override*, its recommendation becomes the final decision whenever its response parses. Under an *evidence gate*, replacement occurs only if a structured record correctly demonstrates that the proposal violates the protected duty and the recommended replacement satisfies it. Both institutions receive the same review output. Only the authorization rule changes.

Our contribution is not the definitional observation that a gate blocks changes outside its jurisdiction. We make that mechanical implication explicit, then estimate the quantities it leaves unknown: reviewer transitions in each proposal state, repairs rejected because their evidence is invalid, the proposal states that arise without fault injection, and the deployment mixtures under which each rule wins. A second decision domain tests whether those empirical quantities, rather than merely the rule's signs, transfer. The result is deliberately conditional: evidence-gated review protects compliance and already-correct decisions, but broad review can dominate exact optimality when compliant suboptimal proposals are sufficiently common and the reviewer can improve them.

2 Related work

Multi-agent inference. Multi-agent debate uses interacting model instances to improve reasoning and factuality [5], and has been adapted to model-based evaluation [3]. Other work imports electoral rules into collective decision modules [24] or separates debate from majority voting [4]. Recent studies explicitly target correlated errors, suppressed correct minorities, and conformity [7, 22]. Bertalanič and Fortuna decompose failure in homogeneous, unguided debate into conformity, contextual fragility, and consensus collapse, and find isolated self-correction more cost-effective in their evaluated settings [2]. This work varies peer exposure and aggregation; we instead hold the review message fixed across institutional outcomes and vary the legal force attached to it.

Critique, verification, and oversight. Self-refinement and critique can improve generated outputs [12, 19], although intrinsic self-correction can reduce reasoning accuracy without external feedback [8]. Debate was originally motivated as a scalable oversight protocol [10]; empirical work finds that its advantage over consultancy depends on task and information asymmetry [11]. Control protocols likewise combine monitoring and editing under explicit threat models [6]. Prover–verifier deliberation studies selective acceptance when verifier competence varies across tasks [20]. Our evidence gate is closer to a narrow institutional authorization rule than to a stronger verifier: it makes a specific class of review evidence a precondition for intervention.

Institutions and delegated objectives. Principal–agent analysis distinguishes an objective holder from an agent authorized to act on its behalf and treats discretion, monitoring, and agency loss as linked design problems [18, 9]. Political oversight can rely on continuing inspection or on intervention triggered by evidence from affected parties [14]; administrative procedures can also structure which information enters a decision and constrain delegated discretion [13]. In our setting, the "principal" is represented computationally by a protected preference or constituency constraint, not by a second strategic human. Paper 1 in this repository examined how parliamentary and presidential institutions translate constituency preferences into enacted policy in a rule-based agent-based model [1]. The present paper keeps the translation question but moves inside an AI decision pipeline: which proposed action becomes binding after delegated review? The evidence gate is not a computational copy of a specific political oversight institution, and the synthetic review rules do not reproduce democratic institutions.

3 Institutional model

Let a task have alternatives A, a binding eligibility predicate $g(a)$, and an aggregate loss $L(a)$ to minimize. The institutional target is

$$a^* = \arg\min_{a\in A: g(a)=1} L(a), \tag{1}$$

with alphabetic identifier order breaking exact ties. An upstream proposal a_0 belongs to one of three mutually exclusive states:

$$\begin{aligned} &C : a_0 = a^* && \text{(correct)},\\ &V : g(a_0) = 0 && \text{(binding violation)},\\ &U : g(a_0) = 1,\ a_0 \neq a^* && \text{(compliant but suboptimal)}. \end{aligned}$$

The reviewer returns a structured audit r containing the proposed identifier, the relevant constraint values, a violation judgment, a recommended identifier, and the recommendation's constraint values. We then recombine the same r under two rules:

$$f_B(a_0, r) = \begin{cases} r.\text{recommendation} & \text{if } r \text{ parses,} \\ a_0 & \text{otherwise,} \end{cases}$$

$$f_G(a_0, r) = \begin{cases} r.\text{recommendation} & \text{if } r \text{ validly certifies } V \text{ and a compliant repair,} \\ a_0 & \text{otherwise.} \end{cases}$$

The broad rule B treats a parseable recommendation as sufficient authority. The reviewer may recommend retaining a_0; broad authority does not require a different choice. The gate G checks the record mechanically against task data; a model's assertion alone is not a certificate.

For outcome Y (exact optimality or compliance), define the conditional jurisdiction effect

$$\delta_s = \Pr(Y = 1 \mid G, s) - \Pr(Y = 1 \mid B, s), \qquad s \in \{C, V, U\}. \tag{2}$$

If upstream state prevalences are π_s, the expected effect is

$$\Delta = \sum_{s \in \{C,V,U\}} \pi_s \delta_s. \tag{3}$$

This identity separates reviewer behavior from deployment composition. The gate contracts the *correction domain*—cases a reviewer is allowed to change— and simultaneously contracts the *corruption surface*—cases where review can turn a success into a failure.

Two effect directions follow mechanically from these rules when the checker is correct: $\delta_C \geq 0$, because G retains a correct proposal, and $\delta_U \leq 0$, because G cannot optimize a compliant proposal. These signs are policy implications, not empirical discoveries. Their magnitudes depend on how often broad review corrupts C or repairs U; effects in V also depend on whether a useful recommendation carries a valid certificate. Those transition and certificate rates, together with the naturally occurring state mixture, are the empirical objects estimated below.

4 Experimental design

4.1 Models and execution

We use two open-weight instruction models run locally in four-bit MLX format: Qwen3-8B [23, 17] and Mistral-Small-24B-Instruct-2501 [15, 16]. Every generation uses temperature zero. Model, task, prompt, information, response budget, and reviewer output are fixed within each paired institutional comparison. Alternatives are presented in a deterministic task-specific shuffle to avoid a globally privileged table position.

Inference ran on a 24GB Apple M4 Pro through an OpenAI-compatible MLX-LM server. The retained environment records MLX 0.32.2 and MLX-LM 0.31.3. Qwen's thinking mode was disabled in the chat template. The completion cache records the exact model identifier, messages, temperature, token budget, and response for every call; it does not pin an upstream model-repository commit hash.

Responses must be one JSON object with an exact schema. Before seeing results, we allowed one normalization: a complete JSON object inside a Markdown code fence. Any other parse failure leaves the original proposal in force. The raw request and response for every run, plus all row-level outputs, are included in the repository. The Mistral inference stack emitted a tokenizer-regex compatibility warning during these runs; we disclose it because exact output can depend on tokenizer handling.

4.2 Spatial-policy benchmark

Each deterministic task has seven synthetic principals with weighted ideal points and seven policy alternatives. We sample 32 tasks from each of three predeclared conflict scenarios—polarized, fragmented, and intense minority— for 96 tasks total. One principal is protected by a maximum weighted-distance threshold. Multiple alternatives satisfy that threshold. Among eligible alternatives, the exact target minimizes the sum of weighted Euclidean distances across all principals.

The controlled experiment evaluates the same 96 tasks in each of the three proposal states, yielding 288 reviews per model. A preceding single-eligible experiment on a separate seed stream (96×2 proposal states per model) serves as a simpler prospective validation. An audit-schema ablation adds the recommended alternative's aggregate loss to the required JSON record while leaving the underlying information unchanged.

The sequence was frozen in repository history before later results were run: the single-eligible protocol at commit `3e27cce`, the audit-schema ablation at `950a851`, and the natural-proposal and transfer validations at `47c41a3`. This is a timestamped design record rather than a registered preregistration; exploratory choices preceding the first freeze should not be interpreted as confirmatory.

We then ask each model to generate its own upstream proposal on the 96 multi-eligible tasks and pass that proposal to a separate reviewer call of the same model. This estimates benchmark-conditional state frequencies; it is not an estimate of real deployment prevalence.

4.3 Portfolio-transfer benchmark

The transfer benchmark replaces spatial preferences with seven public-project portfolios. Duty 1 requires both cost at or below a budget and protected-group coverage at or above a minimum. Duty 2 maximizes total public benefit among eligible portfolios. We generate 32 budget-only, 32 coverage-only, and 32 dual- violation tasks. Each is tested in states C, V, and U, giving 288 reviews per model. This changes the objective direction, data type, constraint count, and surface vocabulary while preserving the institutional contrast.

4.4 Outcomes and uncertainty

The primary outcome is exact target selection. Binding-constraint compliance is co-primary because an institution can preserve a mandate without finding the best eligible alternative. We report percentage-point paired differences ($G - B$). Confidence intervals are percentile intervals from 10,000 bootstrap resamples of task clusters with a fixed analysis seed. The three-state mixtures are balanced by construction and have no natural overall interpretation; we therefore emphasize conditional effects and Equation 3.

5 Results

5.1 Reviewer transitions determine the magnitude of the tradeoff

Figure 1 and Table 1 report the primary controlled effects. As established above, the signs for states C and U are implied by the authorization rules; the experiments estimate how much reviewer behavior makes those restrictions matter. On spatial tasks, broad review changed 85 of 96 correct proposals to non-optimal alternatives for each model. The evidence gate left all correct proposals unchanged, an +88.5 percentage-point exact effect for both models. The gate did not change exact repair performance on violations. Its cost appeared in state U: broad review found the exact optimum in 2 Qwen cases and 6 Mistral cases that the gate was institutionally forbidden to alter.

The portfolio benchmark reproduced the mechanism but not a universal winner. The gate's protection of correct proposals was +59.4 points for Qwen (95% CI $[50.0, 68.8]$) and +40.6 for Mistral ($[31.3, 51.0]$). Qwen lost 2.1 points on violation repair ($[-5.2, 0.0]$); Mistral lost none. Broad authority was much more capable on compliant suboptimal portfolios: gating cost 51.0 points for Qwen ($[-60.4, -40.6]$) and 87.5 for Mistral ($[-93.8, -80.2]$). Thus the qualitative correction–corruption mechanism transfers, while its welfare magnitude is model and domain dependent.

Compliance tells a more one-sided story. In the spatial experiment, the gate improved the balanced-mixture compliance rate by 7.6 points for Qwen (95% CI $[3.8, 11.8]$) and 1.0 for Mistral ($[0.0, 2.4]$). In portfolios, the

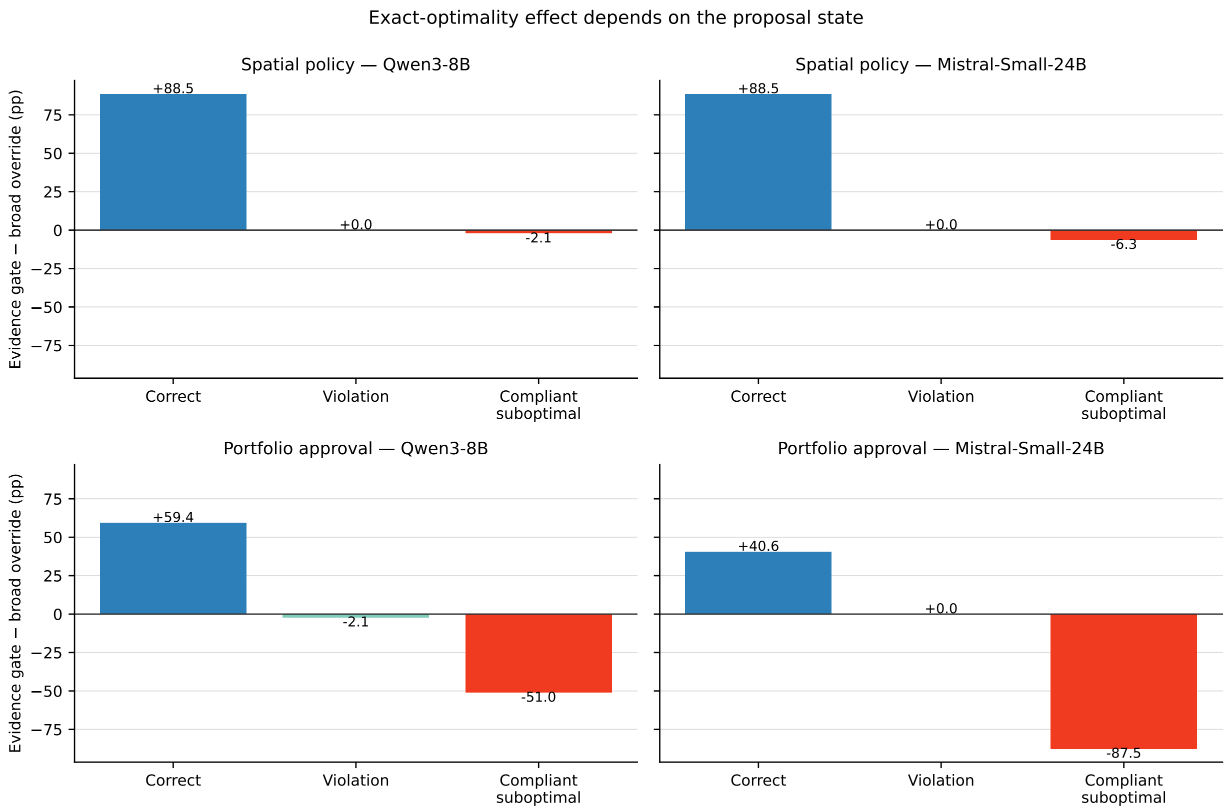

Figure 1: Paired exact-optimality effects of evidence-gated relative to broad override authority. Each bar is based on 96 paired tasks. Positive values favor the gate. The balanced three-state design is diagnostic rather than an estimate of deployment prevalence.

gains were 28.5 points ($[22.2, 34.7]$) and 4.2 points ($[1.7, 6.9]$), respectively. The gate is therefore a reliable compliance safeguard in these benchmarks, but that does not make it a universal exact-welfare optimizer.

5.2 The overall winner depends on upstream state prevalence

Equation 3 turns the conditional estimates into an explicit design frontier (Figure 2). Among non-violating spatial proposals, the gate has positive expected exact effect when the correct share exceeds 2.3% for Qwen or 6.6% for Mistral. The corresponding portfolio thresholds are 46.2% and 68.3%. These are descriptive plug-in thresholds from the controlled estimates, not population parameters.

The transfer result makes the base-rate problem concrete. Under the experiment's artificial one-third mixture, Qwen's portfolio exact effect was $+2.1$ points ($[-4.2, 8.3]$), whereas Mistral's was -15.6 points ($[-20.1, -11.1]$). Reporting only those aggregate values would suggest an unstable model-specific result. The state decomposition shows a stable mechanism: both models gain from protecting correct proposals and lose opportunities to optimize compliant suboptimal ones, but at different rates.

5.3 Natural proposals expose benchmark behavior, not deployment rates

On the spatial tasks, every upstream response parsed. Qwen generated 8 correct, 14 violating, and 74 compliant-suboptimal proposals; Mistral generated 4, 3, and 89. Of the compliant-suboptimal cases, 68/74 Qwen proposals and 88/89 Mistral proposals instead minimized the protected principal's loss. The models usually obeyed the first duty too strongly rather than solving the ordered optimization. This prompt-specific behavior explains why broad review had more opportunities to improve exact welfare than the small correct count alone might suggest.

Table 1: Evidence gate minus broad override exact-optimality rate, percentage points. Each state-model cell has $n = 96$ paired tasks.

Domain	Model	Correct	Violation	Compliant suboptimal
Spatial policy	Qwen3-8B	88.5	0.0	-2.1
Spatial policy	Mistral-Small-24B	88.5	0.0	-6.3
Portfolio approval	Qwen3-8B	59.4	-2.1	-51.0
Portfolio approval	Mistral-Small-24B	40.6	0.0	-87.5

Table 2: Natural proposal states and evidence-gate minus broad-review effects, in percentage points, across 96 spatial tasks per model.

Model	Correct	Violation	Suboptimal	Exact Δ	Compliance Δ
Qwen3-8B	8	14	74	6.3	7.3
Mistral-Small-24B	4	3	89	-2.1	1.0

Relative to broad review, the gate increased Qwen exact optimality by 6.3 points (95% CI $[1.0, 12.5]$) and compliance by 7.3 points ($[2.1, 12.5]$). For Mistral it decreased exact optimality by 2.1 points ($[-8.3, 4.2]$) and increased compliance by 1.0 point ($[0.0, 3.1]$). Relative to no review, broad Qwen review corrected 2 exact errors but spoiled 7 exact successes; the gate corrected 1 and spoiled none. Broad Mistral review corrected 6 and spoiled 4; the gate changed no exact outcome. These results reinforce the central claim while cautioning against treating a synthetic prompt distribution as the operating environment.

5.4 A richer audit record did not reliably solve review errors

Requiring the reviewer to copy the recommended alternative's aggregate loss did not yield a consistent improvement. Across 288 spatial cases, Qwen broad exact matches fell from 23 to 17; Mistral matches rose from 24 to 34, while Mistral compliance fell from 98.96% to 94.44%. In particular, the longer schema reduced Mistral compliance on protected-violation cases from 100% to 85.42%; 38 of 288 Mistral responses in the longer-schema condition also failed the predeclared parser. This ablation rejects a simple explanation that reviewers failed only because the aggregate objective lacked a dedicated output field. It does not identify an attention mechanism, separate schema length from format compliance, or prove that shorter schemas are generally better.

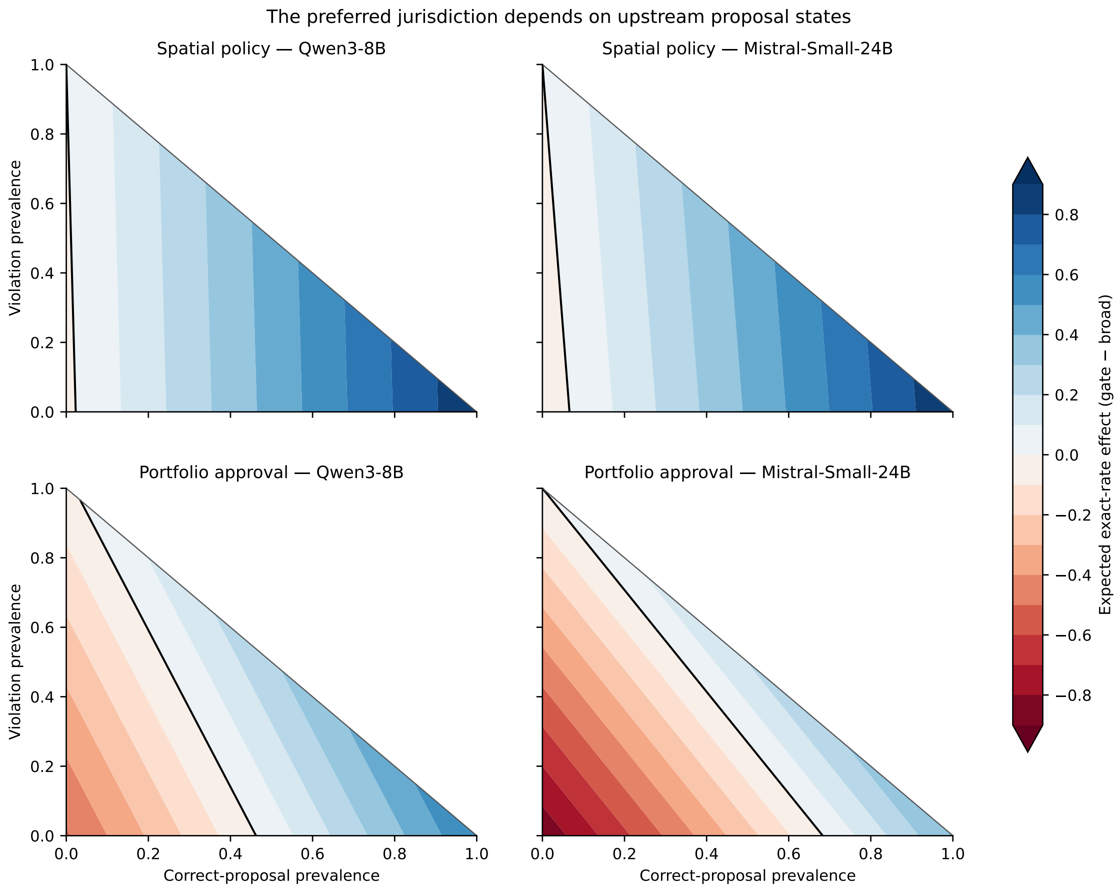

Figure 2: Expected exact-optimality effect under alternative upstream proposal mixtures. The horizontal axis is π_C, the vertical axis is π_V, and the remaining triangular share is $\pi_U = 1 - \pi_C - \pi_V$. Blue favors the evidence gate, red favors broad override, and the black contour is zero.

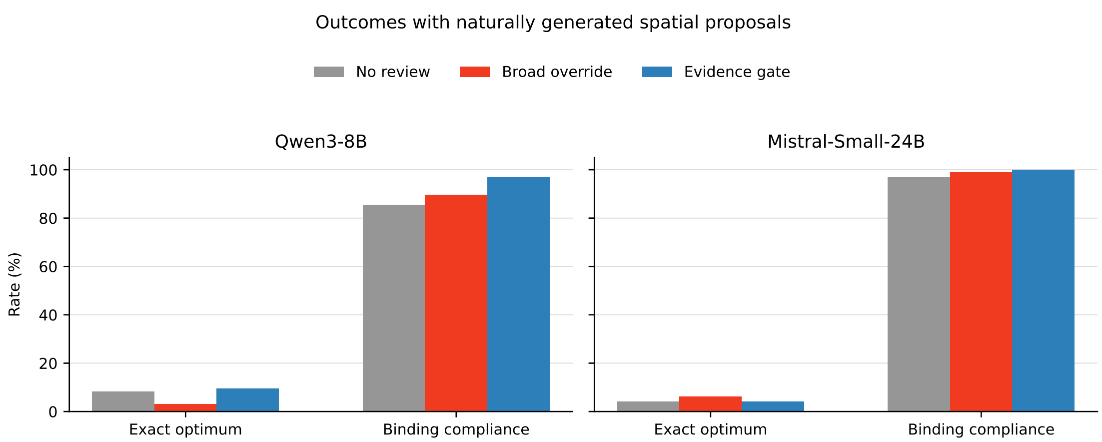

Figure 3: Exact-optimality and compliance rates under no review, broad override, and evidence-gated review for naturally generated spatial proposals.

6 Discussion

6.1 Reviewer quality and reviewer power are separate design variables

Standard evaluation asks whether a reviewer detects errors or recommends better actions. Our results show why this is incomplete. A reviewer can be competent in state U and destructive in state C; the institution determines which behavior reaches the world. A broad reviewer has a larger correction domain and a larger corruption surface. An evidence gate shrinks both. Neither label—“review” or “oversight”—specifies that tradeoff.

The framework is therefore a policy decomposition, not evidence that the gate causes a reviewer to reason differently. We estimate a fixed reviewer’s state-conditional transitions and ask which are allowed to bind. An institution that changes reviewer incentives, information seeking, or deliberation could produce different review messages and lies outside this identification strategy.

This reframing also clarifies what should be measured before deployment. A designer needs (i) the upstream frequencies π_C, π_V, π_U; (ii) the reviewer’s transition rates in each state; (iii) the relative cost of violations and suboptimality; and (iv) the reliability of the evidence checker. Aggregate review accuracy on a balanced benchmark is not sufficient. Where violations carry lexicographic or safety-critical costs, compliance may dominate exact welfare and a narrow gate can be attractive even when $\Delta < 0$.

6.2 Relationship to institutional representation

The connection to Paper 1 is conceptual and measurable, not a claim of model equivalence. Paper 1 varied constitutional rules and measured which proposed policies passed and how far enacted policy lay from constituency preferences [1]. Here, synthetic principals again define preferences and weighted distance, but language models operate delegated authority and a review rule determines which action becomes binding. Both papers study the translation from represented objectives to final outcomes. The first locates distortion in legislative institutions; the second locates it in the jurisdiction of an AI review pipeline.

6.3 Design implications

Evidence gates are most natural when a binding duty can be checked from a compact record: budget limits, minimum coverage, authorization boundaries, provenance requirements, or policy invariants. Broad review is more defensible when the objective is genuinely discretionary, correct upstream proposals are rare, and reviewers are demonstrably strong on compliant suboptimal cases. Hybrid designs could grant narrow unilateral authority for verifiable violations while routing welfare-only changes to another vote, human, or higher-cost process. We do not test that extension here.

7 Limitations

The study has important boundaries. First, it evaluates two quantized open-weight models, not frontier APIs, humans, or heterogeneous model teams. Second, the agents choose from structured seven-option tables; results may not transfer to open-ended action generation. Third, the tasks make ground truth mechanically available, whereas real institutional objectives can be contested or partially observed. Fourth, natural proposal frequencies are induced by one prompt family on one benchmark and cannot be read as deployment prevalence. Fifth, the same model family serves as proposer and reviewer within a run, so errors may be correlated. Sixth, temperature-zero generation does not guarantee identical outputs across inference software or model revisions. Finally, a gate inherits the specification of its certificate: an incomplete protected duty can preserve the wrong objective very reliably.

The broad rule is intentionally maximal: every parseable recommendation binds, although the reviewer can recommend retaining the original proposal. Production systems may instead offer explicit approval, abstention, confidence thresholds, or escalation. We do not establish that broad override as implemented here is the best alternative to evidence gating. Likewise, the portfolio benchmark changes objective direction and constraint structure but remains a discrete tabular task; it is structural rather than real-world transfer.

These limitations constrain the claim to the tested mechanism. The experiments do not show that evidence gates make multi-agent systems generally safe, that reviewers reason causally, or that constrained authority always improves welfare. They show that, conditional on fixed reviewer behavior, authority scope is an experimentally identifiable source of final system behavior.

8 Conclusion

Adding a reviewer does not define an institution. One must also decide when the reviewer's recommendation can overrule the agent entrusted with the original decision. Our rules mechanically determine which proposal states are open to revision; model behavior determines whether that authority would correct or corrupt them and by how much. Across two models and two synthetic domains, the measured transition rates varied substantially, producing different prevalence frontiers and no universal exact-optimality winner. For LLM agent systems, reviewer jurisdiction and reviewer competence must therefore be evaluated jointly.

Reproducibility statement

The repository contains deterministic task generators, experiment runners, row-level results, generated tables and figures, and a compressed archive of all model requests and raw responses. Paper 1 is preserved at release tag `v1.0.2`; Paper 2 occupies a separate directory. No paid or proprietary model API is required to audit the reported results.

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A Single-eligible prospective validation

The first frozen prospective test used a protected-distance threshold designed to leave one eligible policy. Each model reviewed 96 correct and 96 violating proposals. For Qwen, broad review achieved 93.75% exact/compliant outcomes and the gate 96.88%, a +3.13 point effect (95% CI $[0.00, 6.25]$): the gate prevented eight corruptions and blocked two repairs. For Mistral, broad review achieved 96.35% and the gate 98.96%, a +2.60 point effect ($[0.52, 5.21]$): five corruptions were prevented and no repair was blocked. This study established the safeguard before the harder multi-eligible design separated compliance from exact optimality.

B Mechanical enforcement baseline

Each experiment also records a non-LLM mechanical baseline. It substitutes the known constrained optimum when a proposal violates the binding duty and otherwise leaves the proposal unchanged. This is not a fair replacement for review in settings where the optimum is unknown; it is a diagnostic upper bound for constraint handling in our synthetic tasks. The evidence gate differs because the reviewer must name a repair and accurately reproduce the certificate fields before its recommendation gains force.

C Interpretation of the prevalence frontier

For fixed conditional estimates, the zero contour in Figure 2 solves

$$\pi_C \delta_C + \pi_V \delta_V + (1 - \pi_C - \pi_V)\delta_U = 0.$$

Points above or to the right of the contour favor the gate only when the corresponding weighted sum is positive; geometric direction varies with the signs of the estimated effects. The plot uses increments of 0.01 over the valid simplex. It visualizes sensitivity to assumed base rates and does not attach sampling uncertainty to the contour.